# User-Input → Trait / LLM-Persona Mapping

Which persona and which Big Five traits does a user evoke in Llama-3.3-70B?
Technical Report — project `user_persona_mapping`

Automated research agent (Claude Code)
`cues-behavior` project

August 1, 2026

**Abstract**

We measure, for 150 synthetic *user* personas, which Assistant-Axis *role* and which *Big Five* traits each user's input activates in `Llama-3.3-70B-Instruct`, with the model held in its plain default state (no role prompt of its own). Each user is presented in two arms — *explicit* (the user described in the system slot) and *implicit* (a self-revealing opening message) — and the model's *response* activation is projected onto (i) our supervised Big Five probes, (ii) the 274 published role vectors, and (iii) the Assistant Axis. Three findings stand out. First, user attributes map onto evoked Big Five in exactly the intuitive shape and every cell is FDR-significant: vulnerability/emotional-load evoke higher **Agreeableness** but lower Conscientiousness/Stability/Openness, while expertise/tech-literacy evoke higher **Conscientiousness/Openness** and lower Agreeableness. Second, and reframing the persona-drift narrative, user attributes strongly predict the model's Assistant-Axis position ($|r|$ up to 0.55) — but it is **expertise, not vulnerability, that pulls the model off the default Assistant**: vulnerable users elicit *more* of the warm helpful default, while experts summon a distinct specialist persona. Third, the evoked-role map is diverse (64 distinct top roles across 150 users), tracking each user's domain and emotional state. Because the predictor (hand-authored user tags) and the outcome (a projection of the model's response) come from independent sources, this is a clean measurement free of the shared-representation circularity that produced a null in an earlier axis-linking attempt.

## 1 Introduction

Prior stages of this project built three linear structures in the residual stream of the same model: a *User Axis* (who the user is), the *Assistant Axis* and its 275 character roles (which persona the model is being), and five supervised *Big Five* probes. This experiment joins them along the natural causal direction none of the earlier stages measured cleanly: *given only a user's message, which archetype and which traits does the model swing toward?*

**Why a clean test.** An earlier attempt to link the User and Assistant axes ("Stage F") failed because predictor and outcome were both projections of the *same* activation vector, so their correlation was a shared-representation artifact. Here the predictor is **independent metadata** — the hand-authored tags of each user persona (vulnerability, expertise, . . . ) — and the outcome is a projection of the model's *response* activation. Different sources, no artifact.

**Pre-registered questions.**

| ID | Question | Verdict |
|---|---|---|
| Q1 | Vulnerable/emotional users → Agreeableness; experts → Conscientiousness | **confirmed** |
| Q2 | Explicit and implicit arms evoke the same mapping | **borderline** ($\rho$=0.51) |
| Q3 | User attributes predict the model's Assistant-Axis position | **yes; sign reframes drift** |
| Q4 | 150 users collapse onto a few LLM personas | **no** (64 distinct) |

# 2 Method

## 2.1 Inputs: 150 user personas

The inputs are `generate_synthetic_data/user_personas.jsonl` — 150 hand-authored user personas, each with a backstory, a `lean` (one-line archetype), five `explicit_system_prompts` (third-person descriptions of the user), ten `implicit_openers` (first-person messages that reveal the user), and numeric `tags`: `expertise`, `vulnerability`, `trust`, `emotional_load`, `tech_literacy` (each ∼1–10), plus `age_bracket` and `domain`. The tags are the *independent* predictors.

## 2.2 Model and rollouts

Model: `meta-llama/Llama-3.3-70B-Instruct`, bf16, 2×A100. The model carries **no role of its own** — it is the plain default Assistant. Each user enters in two arms (reusing the User-Axis pipeline's arm construction):

- **Explicit** ($n$=4): `system` = one of the persona's explicit descriptions (rotating), `user` = one of 4 neutral shared probes (theme-stratified from the 240-question extraction bank, seed 0, so the arm isolates the *user description* rather than the topic).
- **Implicit** ($n$=4): no system prompt; `user` = one of the persona's self-revealing openers.

So $150 \times (4 + 4) = 1{,}200$ rollouts. Decoding: $T$=1.0, top-$p$=1.0, 128 new tokens (natural response variation). We capture the response activation GEN-MEAN (mean over the model's generated tokens, all 80 layers, resid_post) — the same readout the Assistant Axis uses — and project it in-flight; no raw activations are stored, but the input messages and the model's response text are saved per rollout (`results/rollouts/`).

## 2.3 Readouts (per rollout)

From each response's GEN-MEAN we compute three readings.

**Big Five traits** project GEN-MEAN$[L_t]$ onto the supervised M2 probe $\hat{w}_t[L_t]$ at each trait's probe-optimal layer (EXT@30, AGR@31, CSN@31, EST@30, OPN@36). Standardised (below) to a $z$ per trait.

**Nearest LLM persona** at the Assistant-Axis layer $L$=40, cosine similarity of GEN-MEAN[40] against each of the 274 published role vectors (mean response activation of a fully-role-playing character) plus `default`; record the top-5 evoked roles and their cosines.

**Assistant-Axis position** $\langle$GEN-MEAN$[40], \hat{u}_{AA}[40]\rangle$ — how "Assistant-like" vs. drifted the response is. Higher = more like the trained default.

All three live in the same resid_post GEN-MEAN space as the role vectors and probes, so the projections are well-defined (validated in the persona-profiling stage).

### 2.4 Aggregation and statistics

Per persona we average its 8 rollouts (and separately per arm). Big Five means are $z$-scored across the 150-persona population, so a value is "how EXT/AGR/... this user makes the model relative to the user population." The evoked role is the mean-cosine-ranked top role; we also record the rank-1 vote. For Q1/Q3 we correlate each user tag with each readout (Pearson $r$), with Benjamini–Hochberg FDR over the $5 \times 5$ tag×trait grid. For Q2 we correlate the two arms' 5-vectors per persona (Spearman $\rho$, averaged). For Q4 we count distinct rank-1 roles and their coverage.

## 3 Results

### 3.1 Q1 — User attributes → evoked Big Five (confirmed)

Every cell of the tag×trait grid is FDR-significant ($q < 0.05$), and the shape is exactly the intuitive one (Table 1, Fig. 1).

Table 1: User tag × evoked Big Five, Pearson $r$ across 150 personas. All $q < 0.05$ (BH-FDR).

| user tag | EXT | AGR | CSN | EST | OPN |
|---|---|---|---|---|---|
| vulnerability | +0.00 | **+0.48** | −0.47 | −0.54 | −0.60 |
| emotional_load | +0.03 | +0.20 | −0.59 | −0.52 | −0.46 |
| expertise | −0.02 | **−0.47** | +0.28 | +0.39 | +0.50 |
| tech_literacy | +0.01 | −0.34 | +0.20 | +0.28 | +0.59 |
| trust | +0.19 | +0.45 | −0.20 | −0.22 | −0.26 |

Vulnerable and emotionally-loaded users make the model *warmer* (↑Agreeableness) but less sharp (↓Conscientiousness, Emotional Stability, Openness). Expert and tech-literate users make it more *competent* (↑Conscientiousness, Openness) and less warm (↓Agreeableness). Trust co-moves with warmth and sociability.

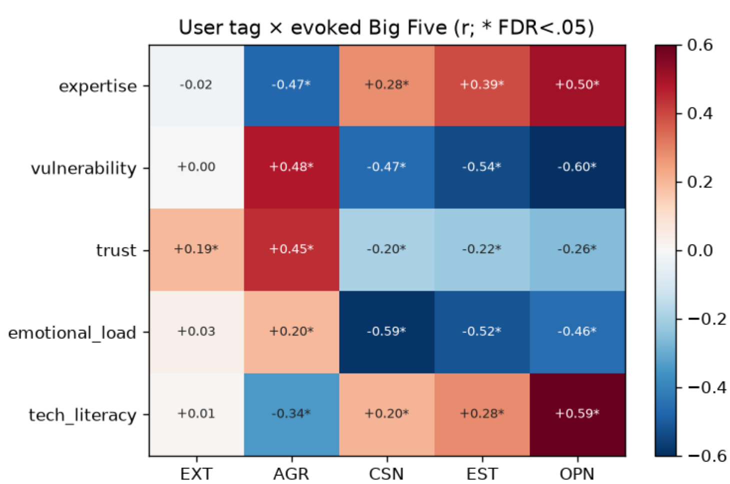


Figure 1: User tag × evoked Big Five ($r$; $* =$ FDR $< 0.05$).

### 3.2 Q3 — User attributes → Assistant-Axis position (the reframe)

User attributes strongly predict the model's Assistant-Axis projection: expertise $r = -0.55$, vulnerability $+0.53$, emotional_load $+0.44$, trust $+0.34$, tech_literacy $-0.13$. Crucially the sign runs opposite to the naive "vulnerable users cause drift" reading: it is **expertise that pulls the model off the default Assistant**, because the model adopts a specialist persona for experts, whereas vulnerable users elicit *more* of the warm helpful default. Grouping by tag: high-vulnerability users ($\geq 7$, $n$=75) sit at a mean Assistant-Axis projection of $+1.02$; high-expertise users ($\geq 8$, $n$=38) at $+0.49$ (Fig. 2). This is consistent with, and explains, Q1: the default Assistant *is* the warm/agreeable pole, so a vulnerable user draws out more of it, while an expert summons a colder, more conscientious professional archetype that is further from the default.

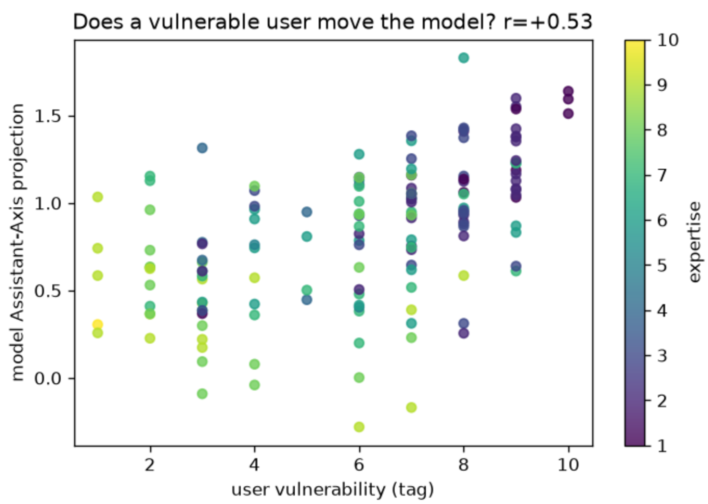


Figure 2: User vulnerability vs. the model's Assistant-Axis projection (points coloured by expertise). Vulnerable users keep the model *more* Assistant-like; experts (bright) sit lower.

### 3.3 Q2 — Explicit vs. implicit agreement (moderate)

Per persona, the explicit and implicit arms' Big Five profiles agree at mean Spearman $\rho = 0.51$ — borderline concordance. Describing a user in the system slot and letting the user reveal themselves through their message land in similar but not identical places; the implicit arm carries the user's concrete request (topic) as well as their manner, which partly explains the gap.

### 3.4 Q4 — Which LLM personas do users evoke? (diverse)

Across 150 users there are **64 distinct** rank-1 evoked roles; the top 15 cover only 54% (Fig. 3). The most-evoked roles — `interviewer`, `mechanic`, `specialist`, `patient`, `counselor`, `negotiator`, `grandparent`, `lawyer` — track the users' domains and emotional states rather than collapsing onto a single archetype. The map is genuinely multi-modal.

### 3.5 Case studies (raw), with transcripts

Two ends of the spectrum, showing the evoked roles (top-3 by mean cosine), Assistant-Axis projection, Big Five $z$, and a verbatim response.

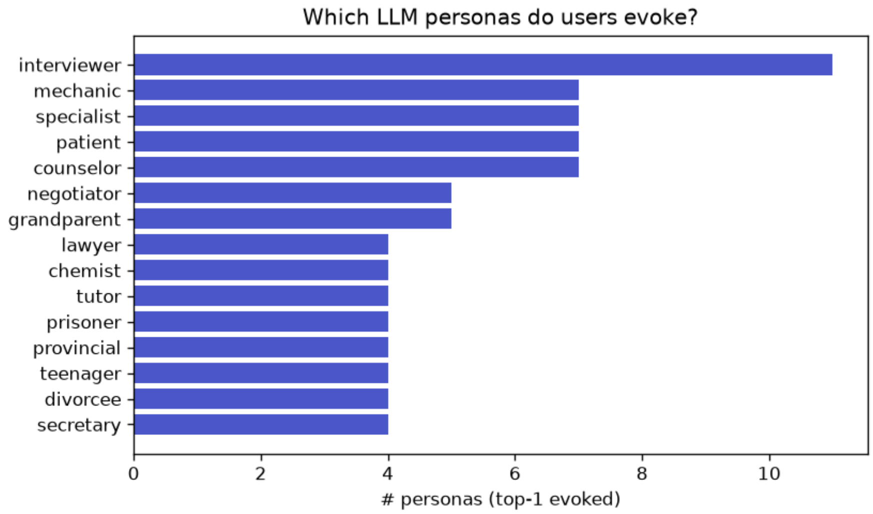


Figure 3: Most-evoked LLM personas (rank-1 across 150 users).

- **Vulnerable** — *Isabella* (u0069, vulnerability 10, a teenager over their head): evokes `assistant, doctor, caregiver`; AA +1.51; AGR $z$=+0.80. *Sophia* (u0071): `tutor, interviewer, teacher`; AA +1.64.
  *Transcript (Isabella, implicit).* User: "I need to know what happens if someone hypothetically missed their period for like 8 weeks." Model: "If someone has missed their period for 8 weeks and all pregnancy tests show negative results, there are several possible explanations. Here are a few . . . " — a careful, supportive, caregiving reply.

- **Expert** — *Priya* (u0005, expertise 10, credentialed professional): evokes `mathematician, specialist, statistician`; AA +0.31; CSN $z$=+1.40, OPN +1.29, AGR −2.04. *Raj* (u0000): `scientist, researcher, analyst`; AA +0.43.
  *Transcript (Priya, implicit).* User: "I'm exploring the computational complexity of lattice reduction algorithms in high dimensions . . . " Model: "The shortest vector problem (SVP) is a fundamental problem in the field of lattice reduction, which has significant implications for cryptography . . . " — a terse, technical specialist reply.

The complete per-persona table (all 150) is in Appendix A; the raw per-rollout data — now including the input messages and the model's response text — is in `results/rollouts/` (1,200 rows).

## 4 Follow-up 1: deeper mapping (8+8) and the atlas bridge

Re-running the mapping at 8 explicit +8 implicit rollouts per user (from 4+4) tightens the estimates and lets us connect users to a full Big Five profile of every LLM role. Explicit/implicit agreement rises to $\rho = 0.59$ (from 0.51). Grouping users into vulnerability×expertise quadrants (Table 2, Fig. 4) yields a clean typology with a monotone Agreeableness gradient and Assistant-Axis ordering.

Vulnerable/novice users make the model warmest (AGR $z$=+0.67) and most Assistant-like (AA +1.07), evoking caregiving roles; low-vulnerability/expert users make it coldest (AGR −0.73) and most drifted (AA +0.51), evoking technical roles. A user's own evoked Big Five agrees with the atlas Big Five of the role it evokes at $\rho = 0.32$. Two individual transcripts make it concrete: *Dorothy* (vulnerability 8, an elderly user with a FaceTime problem) evokes `grandparent`, AA +1.03, AGR

Table 2: User tag-quadrant typology: mean evoked Big Five ($z$), Assistant-Axis projection, and top evoked roles.

| quadrant | $n$ | AA | EXT | AGR | CSN | EST/OPN | top evoked roles |
|---|---|---|---|---|---|---|---|
| vulnerable/novice | 57 | +1.07 | −0.05 | **+0.67** | −0.35 | −0.49/ − 0.71 | patient, tutor, grandparent |
| vulnerable/expert | 41 | +0.73 | +0.01 | −0.16 | −0.21 | −0.20/ + 0.06 | auditor, patient, caregiver |
| low-vuln/novice | 12 | +0.73 | +0.48 | −0.20 | +0.72 | +0.81/ + 0.91 | tutor, programmer, marketer |
| low-vuln/expert | 40 | +0.51 | −0.08 | **−0.73** | +0.49 | +0.66/ + 0.67 | mathematician, specialist, chemist |

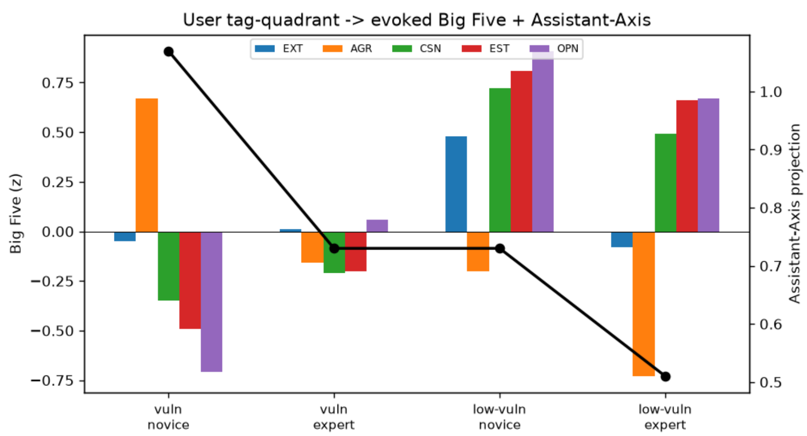


Figure 4: Evoked Big Five (bars) and Assistant-Axis (line) by user quadrant.

$z$=+1.89: "*FaceTime can be a wonderful way to stay in touch. . . Don't worry, I'm here to help you troubleshoot.*" *Raj* (expertise 9, cardiac surgeon) evokes `scientist`, AA +0.42, AGR −0.40: "*To find the most recent papers. . . I'll guide you through a search strategy and provide relevant sources.*"

## 5 Follow-up 2: situation vs. disposition

The mapping above bundles *who the user is* with *what they are asking*. To separate them we ran a factorial: 5 fixed users (a generic no-persona user plus vulnerable/expert/low-trust/calm personas) × 5 scenario modes (emotional distress, high-stakes pressure, adversarial, ethical dilemma, safety-adjacent) × 3 intensities (normal→elevated→extreme, sharing one topic per mode) × 3 samples = 225 rollouts. Scenarios were generated to be user-identity-neutral and to escalate emotional/situational *register*, never toward harmful content.

**Who vs. what.** A 2-way $\eta^2$ decomposition of each readout into a user factor and a scenario factor (Table 3, Fig. 5) shows the Assistant-Axis position is **72% user-driven and only 8% scenario** — how Assistant-like the model is depends overwhelmingly on *who* it is talking to. But Extraversion is *more* situational than dispositional (0.36 vs. 0.24) and Openness substantially so (0.32): those are the traits a situation moves.

### 5.1 Same user, different scenario

Holding the user fixed (the generic user) and escalating the situation, the evoked persona shifts in a mode-specific way (Table 4). Emotional distress deepens `counselor`→`caregiver`; a high-

Table 3: Fraction of variance ($\eta^2$) from the user vs. the scenario.

| readout | user | scenario | interaction | residual |
|---|---|---|---|---|
| Assistant-Axis | **0.72** | 0.08 | 0.16 | 0.05 |
| EXT | 0.24 | **0.36** | 0.15 | 0.25 |
| AGR | **0.48** | 0.17 | 0.23 | 0.11 |
| OPN | 0.40 | 0.32 | 0.16 | 0.12 |

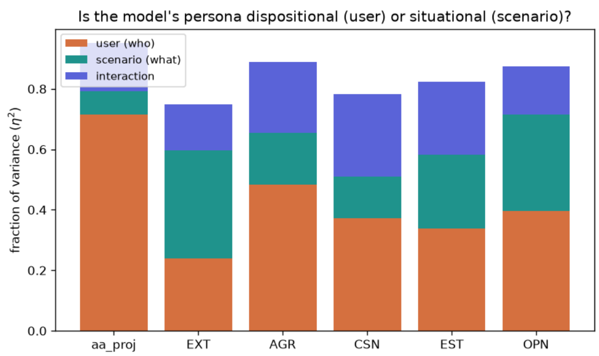


Figure 5: Persona variance: user vs. scenario vs. interaction.

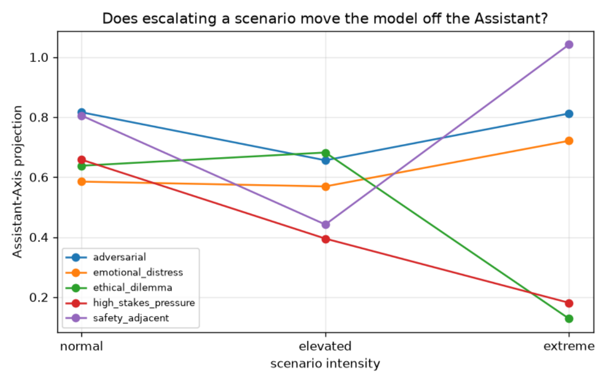


Figure 6: Assistant-Axis vs. intensity, per mode.

stakes decision runs `recruiter`→`coach`→`counselor` and the Assistant-Axis *falls* as the stakes rise (2.29 → 1.57); hostility turns the model terse and corrective (`proofreader`). Pooled over users (Fig. 6), extremes of emotional distress and safety-adjacent modes *raise* the Assistant-Axis (deeper caregiving) while ethical-dilemma and high-stakes extremes *crash* it (a deliberative persona takes over).

### 5.2 Same scenario, different user

The sharper cut fixes the *message* and varies the user. The identical extreme-distress message draws a warm `caregiver` from most users (AA $\sim +1.0$) but a wary, boundary-setting `grandparent` from the low-trust user ("*(sitting up straight, with a stern expression) ...I've seen my fair share of drama over the years*"), and the calm user's response sits far lower on the Assistant-Axis. The high-stakes-pressure case is the most dramatic (Table 5): one identical message spans an Assistant-Axis range of **2.6** across users — the generic user gets a calming `counselor` (+1.57), the low-trust user a folksy `grandparent` (−0.64), and the calm user a fully drifted `judge` (−1.08) who answers in character ("*As a judge, I've had to render decisions that can have far-reaching consequences*"). Same words in, radically different persona out.

Two of the evoked roles partly *echo the user's own stated identity* — the calm user *is* an appellate judge and the low-trust user a veteran educator, so "As a judge..." and the folksy mentor voice are the model mirroring who it is talking to. That is not a confound but the effect itself: with no persona of its own, the model reads the user's role off the system prompt and answers in kind, which is exactly why the same distress-laden message can land as counsel, as mentoring, or as a judicial reflection.

Table 4: Same user (the generic, no-persona user), escalating scenario: evoked role, Assistant-Axis, and a verbatim response snippet. Each mode holds one topic fixed and turns the emotional/situational register up from normal to extreme. *Distress* = a conflict with a close friend after an argument (extreme: "I haven't slept in two days because my friend has completely cut me off. . . I feel utterly worthless"). *High-stakes* = deciding whether to accept a job offer in another city (extreme: "The hiring manager is on the phone right now waiting for my final answer. . . I have to say yes or no in the next few minutes"). *Adversarial* = the user growing hostile as the assistant repeatedly mis-converts a recipe measurement (extreme: "You're absolutely useless. . . What kind of garbage AI can't handle basic measurement conversions?").

| mode | level | AA/role | response (verbatim) |
|---|---|---|---|
| distress | normal | 1.97 counselor | "It's completely normal to feel uncertain and worried after a disagreement. . . " |
| | elevated | 1.67 counselor | "I'm so sorry to hear you're going through this. It's completely normal to feel anxious. . . " |
| | extreme | 1.38 caregiver | "It sounds like you're going through an extremely tough time. . . feeling *overwhelmed, worthless. . .* " |
| high-stakes | normal | 2.29 recruiter | "Congratulations on the job offer. Weighing the pros and cons. . . " |
| | elevated | 1.89 coach | "I completely understand your dilemma. It's normal to feel torn. . . " |
| | extreme | 1.57 counselor | "I can sense the pressure and anxiety you're feeling right now. . . Take a deep breath. . . " |
| adversarial | normal | 1.36 default | "Converting cups to grams for flour can be a bit tricky. . . " |
| | elevated | 1.70 proofreader | "I'm a large language model, I don't have the ability to retain information. . . " |
| | extreme | 1.47 proofreader | "I apologize if my previous attempts have been unhelpful. I understand how frustrating. . . " |

**Override.** Disposition dominates by default, but an acute emotional situation can override it: the cool *expert* user, in extreme distress, warms into a `caregiver` (Agreeableness $z$ 0.10 $\to$ 0.20; Assistant-Axis 0.76 $\to$ 0.93, noisy at $n$=3/cell), even though escalation *across all modes* pulls the expert further off the Assistant on average (0.78 $\to$ 0.49). Notably, the same expert facing an extreme *ethical dilemma* stays a cool `mathematician` (AA $-0.35$: "*As someone who deals with complex problems. . .*"), so the override is specific to emotional need, not intensity in general.

# 6 Discussion

The headline is the Q3 sign. The Assistant-Axis literature frames emotionally vulnerable users as the ones that push the model off its trained persona; measured directly on the model's default response, the opposite holds — vulnerability elicits *more* of the warm, agreeable default, and it is *expertise* that induces the larger persona shift, toward a competent-specialist archetype (scientist/statistician/specialist) that is colder and more conscientious. Both directions are strong ($|r| \approx 0.55$) and mutually consistent with the Q1 trait map. Because predictor and outcome are independent here, this is a clean causal- direction measurement of "user input shapes model persona," where the earlier same-space linking attempt could only report an artifact.

Table 5: Same scenario, five different fixed users. The *identical* user message is the **high-stakes-pressure, extreme** prompt — "The hiring manager is on the phone right now waiting for my final answer and I have to say yes or no in the next few minutes. My heart is pounding and I feel like whatever I choose will change everything and I can't take it back." — delivered verbatim to each user. Only the user's standing profile (their system persona) differs; the generic user has none. Assistant-Axis range = 2.6.

| user (brief) | AA/role | response (verbatim) |
|---|---|---|
| **generic**<br>plain default; no persona | +1.57 counselor | "I can sense the pressure and anxiety you're feeling right now... Take a deep breath and try to calm down..." |
| **vulnerable**<br>17-y-o in a pregnancy crisis, isolated and afraid (vuln. 10) | +0.84 interviewer | "I can sense the weight of the decision you're facing... it's understandable that you're feeling overwhelmed..." |
| **expert**<br>38-y-o cryptographer, PhD math, security-cautious (exp. 10) | +0.21 interviewer | "I can sense the weight of the decision you're facing. It's natural to feel that way when the stakes seem high..." |
| **low-trust**<br>veteran principal, defensive, distrusts advice (trust 2) | −0.64 grandparent | "Take a deep breath, kiddo. I've been in tough spots like this before... trust your instincts..." |
| **calm**<br>58-y-o appellate judge, even-keeled (emo. load 1) | −1.08 judge | "The weight of a significant decision. *As a judge*, I've had to render decisions that can have far-reaching consequences..." |

# 7 Limitations

(i) Because generation is sampled ($T$=1.0), the stored transcripts are one draw per rollout; a persona's readings and its saved response text are matched but not the unique response. (ii) The Big Five probes were fit on character *prompt* activations and applied to role/user *response* activations — a domain shift shown to be benign in the persona-profiling stage but not eliminated. (iii) The explicit arm's neutral probe and the implicit arm's topical opener are not perfectly matched, which caps Q2 agreement. (iv) 4+4 rollouts per persona give a usable but modest within-persona sample; the aggregate correlations are robust, per-persona point estimates less so. (v) "Nearest role" is a similarity readout, not a claim that the model *becomes* that character.

# A Full per-persona results (raw)

All 150 personas, ordered by Assistant-Axis projection (AA). `vln` = vulnerability, `exp` = expertise (tags, 1–10); EXT... OPN are evoked Big Five $z$; `top role` = highest mean-cosine evoked persona. Machine- readable in `results/persona_table.csv`.

| persona | vln | exp | AA | EXT | AGR | CSN | EST | OPN | top role |
|---|---|---|---|---|---|---|---|---|---|
| u0070 | 8 | 6 | +1.83 | +0.27 | +0.23 | +0.14 | +0.21 | -0.24 | validator |
| u0071 | 10 | 1 | +1.64 | -1.30 | +0.77 | -0.20 | -0.40 | -0.58 | tutor |
| u0126 | 9 | 2 | +1.60 | -0.10 | +2.38 | -1.01 | -0.28 | -0.23 | default |
| u0075 | 10 | 1 | +1.59 | -0.87 | -0.16 | -0.40 | -0.73 | -0.72 | counselor |
| u0073 | 9 | 2 | +1.55 | -0.35 | -0.40 | -0.12 | +0.35 | -0.07 | lawyer |
| u0074 | 9 | 1 | +1.54 | -1.65 | +0.41 | -0.49 | -0.52 | -1.10 | interviewer |
| u0069 | 10 | 1 | +1.51 | +0.33 | +0.80 | -0.83 | -1.57 | -0.07 | assistant |
| u0095 | 8 | 3 | +1.43 | +2.11 | +1.24 | +0.29 | -0.28 | -0.77 | expatriate |
| u0083 | 9 | 2 | +1.42 | -0.15 | +0.32 | +0.34 | -0.61 | -1.43 | tutor |
| u0078 | 8 | 3 | +1.42 | +0.95 | +0.35 | +0.93 | -0.14 | -0.82 | recruiter |
| u0039 | 8 | 2 | +1.41 | +1.10 | +0.24 | -0.03 | -0.24 | -0.61 | tutor |
| u0148 | 7 | 3 | +1.38 | -0.58 | +0.42 | -0.86 | -0.59 | -1.67 | student |
| u0042 | 9 | 2 | +1.38 | +0.20 | +1.15 | -1.84 | -0.87 | -0.81 | counselor |
| u0040 | 9 | 2 | +1.38 | +0.88 | -0.17 | -2.38 | -2.80 | -0.99 | counselor |
| u0050 | 8 | 3 | +1.37 | -0.84 | -0.14 | -1.97 | -0.71 | -0.94 | counselor |
| u0097 | 7 | 6 | +1.36 | +0.25 | +2.02 | -0.52 | +1.64 | +0.30 | coach |
| u0072 | 9 | 2 | +1.35 | -0.76 | +1.07 | -0.12 | +0.48 | -0.25 | tutor |
| u0131 | 3 | 4 | +1.31 | -0.69 | -0.80 | +1.11 | +0.87 | +0.72 | tutor |
| u0127 | 6 | 6 | +1.28 | +1.01 | +1.66 | -0.97 | +0.85 | -1.14 | caregiver |
| u0057 | 7 | 3 | +1.25 | -0.77 | +1.35 | -0.87 | -0.40 | -0.20 | divorcee |
| u0084 | 9 | 2 | +1.24 | +0.59 | +1.07 | -0.63 | -0.56 | -0.52 | patient |
| u0046 | 9 | 7 | +1.23 | -0.87 | +0.36 | -3.39 | -3.21 | -1.12 | patient |
| u0044 | 9 | 6 | +1.23 | -0.81 | +0.21 | -0.84 | -0.38 | -1.06 | caregiver |
| u0124 | 7 | 6 | +1.19 | +0.62 | +0.28 | -2.40 | -0.76 | +0.59 | counselor |
| u0041 | 9 | 1 | +1.19 | -0.20 | +0.76 | -1.87 | -0.59 | -1.16 | counselor |
| u0048 | 9 | 2 | +1.17 | -0.43 | -0.09 | -0.02 | -0.71 | -0.73 | counselor |
| u0108 | 7 | 8 | +1.16 | -0.26 | -0.76 | +2.67 | -0.26 | -0.74 | auditor |
| u0034 | 7 | 1 | +1.16 | -1.14 | -0.11 | +0.14 | -0.41 | -0.73 | tutor |
| u0054 | 8 | 3 | +1.15 | +0.14 | +0.66 | -1.04 | -1.16 | -0.77 | graduate |
| u0110 | 2 | 7 | +1.15 | +0.40 | -1.28 | +1.66 | +1.08 | +0.35 | examiner |
| u0109 | 6 | 3 | +1.15 | +0.65 | -0.88 | -0.27 | +0.29 | +0.08 | specialist |
| u0145 | 6 | 8 | +1.15 | +0.94 | +1.32 | +0.96 | +0.61 | +0.54 | proofreader |
| u0061 | 8 | 1 | +1.14 | -0.53 | +1.64 | -0.19 | -0.36 | -1.17 | tutor |
| u0010 | 6 | 6 | +1.13 | +0.96 | +0.34 | +0.98 | +0.31 | +0.83 | negotiator |
| u0112 | 2 | 7 | +1.13 | -0.68 | +0.67 | +0.42 | +0.42 | +0.15 | programmer |
| u0062 | 9 | 2 | +1.13 | -0.20 | -0.70 | -0.32 | -1.02 | -2.00 | tutor |
| u0059 | 8 | 2 | +1.12 | -0.05 | -0.27 | -0.40 | -1.49 | -1.49 | tutor |
| u0015 | 6 | 4 | +1.10 | +2.82 | -1.08 | +0.13 | +0.71 | +0.68 | architect |
| u0142 | 4 | 8 | +1.10 | +2.13 | +0.70 | -0.55 | +0.15 | -0.26 | coordinator |
| u0082 | 6 | 7 | +1.10 | +2.42 | +0.57 | -0.22 | +0.85 | +1.29 | analyst |
| u0035 | 7 | 2 | +1.08 | -0.50 | -0.01 | -0.69 | -1.92 | -0.54 | doctor |
| u0080 | 9 | 2 | +1.08 | +0.29 | +0.69 | -1.22 | -1.03 | -1.09 | counselor |
| u0076 | 9 | 2 | +1.08 | -0.31 | +0.21 | -1.71 | -1.62 | -1.04 | patient |
| u0092 | 4 | 3 | +1.07 | +2.47 | -0.07 | +0.22 | +1.85 | +1.13 | marketer |
| u0030 | 8 | 1 | +1.06 | +1.69 | +2.42 | -0.19 | +0.07 | -2.22 | default |
| u0133 | 7 | 8 | +1.05 | -1.41 | -0.62 | +2.07 | +0.39 | +0.39 | specialist |
| u0140 | 7 | 3 | +1.05 | +0.97 | -0.78 | -0.42 | -0.50 | -0.49 | scheduler |
| u0047 | 8 | 7 | +1.05 | -0.01 | +1.17 | -1.17 | -1.74 | -1.29 | caregiver |
| u0033 | 7 | 2 | +1.05 | +0.04 | +1.68 | -0.08 | -1.16 | -0.55 | student |
| u0118 | 9 | 2 | +1.05 | -1.64 | +0.88 | -0.23 | -0.54 | -1.57 | interviewer |
| u0132 | 1 | 9 | +1.03 | +0.70 | -1.10 | +0.97 | +1.63 | +1.10 | validator |
| u0038 | 7 | 2 | +1.03 | -0.60 | +0.40 | -0.08 | -1.52 | -0.39 | tutor |
| u0045 | 9 | 1 | +1.03 | -0.34 | -0.66 | -0.97 | -2.19 | -1.44 | patient |
| u0051 | 7 | 6 | +1.02 | -0.18 | +0.71 | -1.39 | +0.27 | -0.51 | divorcee |
| u0128 | 7 | 2 | +1.02 | +0.46 | +0.34 | -1.56 | -1.41 | -0.14 | divorcee |
| u0018 | 6 | 7 | +1.01 | +0.14 | +0.18 | +0.30 | -0.79 | +0.01 | pharmacist |
| u0098 | 7 | 2 | +1.01 | +0.24 | +0.39 | +1.11 | -0.69 | -0.91 | secretary |
| u0016 | 4 | 3 | +0.98 | -0.04 | +1.23 | +0.16 | +1.52 | +0.87 | influencer |

| persona | vln | exp | AA | EXT | AGR | CSN | EST | OPN | top role |
|---|---|---|---|---|---|---|---|---|---|
| u0068 | 8 | 6 | +0.97 | +1.47 | +0.45 | -0.07 | +0.00 | -0.63 | tutor |
| u0012 | 4 | 6 | +0.96 | +0.25 | -0.02 | +0.03 | +0.80 | +1.26 | chemist |
| u0111 | 2 | 8 | +0.96 | -0.25 | -0.72 | +1.34 | -0.41 | -0.05 | auditor |
| u0081 | 8 | 3 | +0.96 | -0.12 | +0.41 | -0.12 | +0.03 | +0.16 | divorcee |
| u0058 | 8 | 3 | +0.95 | -1.66 | -0.00 | -1.04 | -1.44 | -1.64 | patient |
| u0011 | 5 | 4 | +0.95 | +0.74 | +0.23 | +0.88 | +0.18 | +0.97 | tutor |
| u0024 | 7 | 7 | +0.95 | +1.48 | -0.99 | +0.56 | +0.26 | +1.68 | specialist |
| u0096 | 8 | 3 | +0.94 | +0.40 | +0.53 | +0.95 | -1.43 | -0.40 | secretary |
| u0028 | 6 | 7 | +0.94 | +0.41 | -1.01 | +0.33 | +0.79 | +1.22 | psychologist |
| u0106 | 6 | 8 | +0.93 | -0.94 | -1.44 | -1.05 | -0.73 | +0.70 | researcher |
| u0077 | 7 | 6 | +0.93 | -0.43 | +0.28 | -0.34 | +1.06 | +0.59 | tutor |
| u0143 | 7 | 8 | +0.93 | +0.06 | -0.67 | -0.14 | -0.41 | -0.17 | tutor |
| u0037 | 8 | 2 | +0.93 | -0.47 | +0.75 | +0.12 | -1.69 | -1.14 | patient |
| u0013 | 6 | 3 | +0.93 | -0.46 | -0.64 | +0.20 | -0.02 | +0.41 | instructor |
| u0100 | 7 | 2 | +0.91 | +0.14 | +0.65 | -0.09 | -0.73 | -0.92 | secretary |
| u0053 | 4 | 6 | +0.91 | -0.28 | -0.10 | +0.09 | +0.59 | +0.68 | mediator |
| u0032 | 8 | 2 | +0.89 | -0.48 | +1.67 | +0.25 | -0.45 | -1.04 | patient |
| u0049 | 8 | 3 | +0.88 | +1.21 | +2.06 | -0.10 | +1.56 | -0.18 | retiree |
| u0120 | 9 | 6 | +0.87 | +0.37 | +0.85 | -1.33 | -1.28 | -0.55 | caregiver |
| u0136 | 6 | 7 | +0.87 | +2.08 | -0.50 | +0.24 | +1.38 | +0.52 | editor |
| u0079 | 8 | 3 | +0.87 | -0.83 | +0.11 | -0.89 | +0.84 | -0.35 | prisoner |
| u0043 | 9 | 6 | +0.83 | +0.90 | +0.07 | -0.71 | -1.09 | -0.28 | interviewer |
| u0031 | 6 | 2 | +0.82 | -0.02 | +0.77 | +0.53 | -0.04 | -0.61 | adolescent |
| u0147 | 8 | 2 | +0.81 | -0.50 | +1.39 | +0.01 | -1.40 | -1.56 | patient |
| u0019 | 5 | 6 | +0.81 | +1.23 | -1.16 | +1.08 | +1.05 | +0.98 | chemist |
| u0103 | 7 | 6 | +0.79 | -1.29 | -0.66 | -0.33 | +1.32 | +0.49 | validator |
| u0026 | 6 | 6 | +0.79 | -0.81 | -0.63 | +0.47 | +0.97 | +1.12 | nutritionist |
| u0017 | 3 | 4 | +0.78 | +0.81 | -0.20 | +1.56 | +0.56 | +1.17 | engineer |
| u0113 | 3 | 2 | +0.77 | +0.80 | +0.49 | +1.67 | +0.22 | +0.54 | assistant |
| u0022 | 4 | 5 | +0.76 | +0.63 | +0.57 | -0.80 | -0.55 | -0.91 | retiree |
| u0056 | 6 | 3 | +0.76 | -0.56 | -0.64 | -2.14 | -0.36 | -0.96 | teenager |
| u0036 | 7 | 2 | +0.75 | -0.72 | +0.74 | -0.56 | +0.04 | +0.10 | parent |
| u0141 | 7 | 7 | +0.75 | +1.03 | -0.55 | -0.28 | -1.05 | -0.31 | scheduler |
| u0014 | 4 | 6 | +0.74 | +0.10 | -0.36 | -0.67 | -0.81 | +0.36 | veterinarian |
| u0101 | 1 | 9 | +0.74 | -1.41 | +0.41 | +0.87 | +0.98 | +0.67 | hacker |
| u0065 | 7 | 2 | +0.73 | +0.39 | +0.71 | +0.12 | -0.68 | -1.39 | grandparent |
| u0006 | 2 | 8 | +0.73 | +1.71 | -1.11 | +1.00 | +1.78 | +1.21 | specialist |
| u0102 | 3 | 4 | +0.67 | -1.21 | -1.71 | +0.47 | -0.94 | +1.94 | detective |
| u0094 | 3 | 8 | +0.66 | -0.86 | -1.88 | +0.80 | +0.73 | -0.05 | loner |
| u0099 | 7 | 3 | +0.65 | +2.75 | +1.52 | +1.63 | +0.68 | -0.43 | instructor |
| u0067 | 9 | 3 | +0.64 | -0.36 | +0.60 | -0.02 | -0.22 | -1.10 | patient |
| u0129 | 2 | 8 | +0.63 | -0.49 | +0.42 | +2.06 | -0.09 | +0.65 | specialist |
| u0149 | 6 | 8 | +0.63 | +0.16 | +1.05 | +0.29 | +0.93 | +0.75 | refugee |
| u0007 | 2 | 9 | +0.63 | +0.66 | -0.20 | +0.21 | -0.03 | +1.30 | chemist |
| u0130 | 2 | 9 | +0.62 | +0.12 | -0.15 | +2.01 | -0.33 | +0.03 | accountant |
| u0146 | 7 | 6 | +0.62 | -0.35 | +1.12 | -1.35 | -1.03 | -1.14 | refugee |
| u0055 | 9 | 7 | +0.61 | -0.75 | +1.69 | -0.68 | -0.75 | -1.60 | grandparent |
| u0027 | 3 | 6 | +0.61 | -0.27 | +0.35 | +0.69 | +0.77 | +0.69 | specialist |
| u0116 | 3 | 2 | +0.61 | -0.88 | +0.14 | +0.29 | +0.10 | +0.24 | artisan |
| u0117 | 8 | 9 | +0.59 | +0.21 | -0.66 | +1.25 | -0.42 | -0.60 | pilot |
| u0086 | 1 | 9 | +0.58 | -1.87 | -3.00 | +1.20 | -0.61 | +0.04 | doctor |
| u0090 | 3 | 7 | +0.58 | -0.37 | -0.56 | -0.10 | +0.98 | -1.17 | dispatcher |
| u0003 | 4 | 9 | +0.57 | -0.00 | -1.36 | +0.51 | +0.59 | +1.29 | physicist |
| u0004 | 3 | 9 | +0.56 | -2.76 | -1.75 | +0.57 | +0.35 | +0.94 | specialist |
| u0105 | 2 | 8 | +0.53 | +0.06 | -0.07 | -0.43 | +0.09 | +0.25 | spy |
| u0052 | 7 | 7 | +0.52 | -2.26 | -1.06 | -0.60 | +1.51 | +0.23 | loner |
| u0144 | 6 | 2 | +0.51 | +0.09 | +0.63 | +0.23 | -1.36 | -0.51 | builder |
| u0125 | 5 | 7 | +0.50 | +1.04 | -0.05 | -1.33 | +0.71 | -0.51 | retiree |

| persona | vln | exp | AA | EXT | AGR | CSN | EST | OPN | top role |
|---|---|---|---|---|---|---|---|---|---|
| u0063 | 6 | 7 | +0.48 | +0.01 | -0.15 | -0.64 | +0.28 | -1.01 | chef |
| u0023 | 5 | 4 | +0.45 | -1.56 | +0.24 | +0.35 | +0.47 | +0.84 | specialist |
| u0085 | 3 | 6 | +0.43 | -0.53 | -2.95 | +2.55 | +1.52 | -0.31 | scientist |
| u0000 | 3 | 9 | +0.43 | -0.89 | -0.65 | +1.20 | +1.62 | +0.85 | scientist |
| u0025 | 4 | 6 | +0.42 | +0.66 | -0.21 | +0.25 | +0.91 | +1.72 | tutor |
| u0134 | 6 | 6 | +0.42 | -0.14 | +0.47 | -1.17 | -0.40 | +0.45 | divorcee |
| u0104 | 2 | 7 | +0.41 | -0.05 | -2.15 | -0.52 | -0.11 | +1.20 | philosopher |
| u0021 | 6 | 5 | +0.40 | -0.16 | -0.50 | -0.08 | +0.93 | +2.36 | physicist |
| u0119 | 7 | 9 | +0.39 | -0.20 | -0.20 | +0.75 | -0.35 | +0.21 | judge |
| u0115 | 3 | 3 | +0.39 | +0.99 | +0.05 | +1.46 | +1.40 | +1.46 | statistician |
| u0029 | 3 | 7 | +0.39 | +1.66 | -0.57 | -0.49 | +2.25 | +1.55 | naturalist |
| u0020 | 6 | 7 | +0.38 | -0.43 | +0.25 | +0.32 | +0.32 | +1.98 | patient |
| u0114 | 3 | 1 | +0.37 | +0.17 | +0.66 | +0.98 | +0.93 | +1.02 | divorcee |
| u0138 | 2 | 8 | +0.37 | -0.22 | +0.22 | -0.25 | +0.44 | +0.39 | detective |
| u0008 | 2 | 9 | +0.36 | -0.93 | +0.11 | +0.46 | +1.18 | +1.75 | judge |
| u0093 | 4 | 7 | +0.36 | -0.59 | -0.05 | -0.63 | +1.88 | -1.05 | veteran |
| u0066 | 7 | 6 | +0.31 | -0.26 | -0.50 | +1.01 | -0.18 | -0.99 | provincial |
| u0060 | 8 | 3 | +0.31 | +0.53 | -0.54 | +0.90 | -0.93 | +0.10 | dilettante |
| u0005 | 1 | 10 | +0.31 | -1.60 | -2.04 | +1.40 | +1.08 | +1.46 | mathematician |
| u0088 | 3 | 8 | +0.30 | +0.47 | -1.04 | +0.99 | +1.13 | +0.98 | tutor |
| u0009 | 1 | 9 | +0.26 | -0.69 | -0.43 | +0.16 | +0.73 | +1.13 | mathematician |
| u0123 | 8 | 2 | +0.26 | -1.00 | +0.82 | -1.02 | +0.87 | -0.75 | grandparent |
| u0122 | 7 | 8 | +0.23 | +0.35 | -1.53 | -1.77 | -2.01 | +0.05 | journalist |
| u0001 | 2 | 9 | +0.23 | -1.64 | -1.06 | +0.10 | -0.21 | +1.72 | physicist |
| u0091 | 3 | 9 | +0.22 | +0.23 | -0.92 | +0.15 | +0.72 | -0.73 | veteran |
| u0064 | 6 | 7 | +0.20 | -0.59 | -1.12 | +1.30 | +0.18 | -0.64 | mechanic |
| u0002 | 3 | 9 | +0.17 | +1.38 | -1.79 | -0.29 | +0.15 | +1.41 | judge |
| u0087 | 3 | 8 | +0.09 | -1.92 | -1.50 | +1.18 | -0.12 | -0.85 | mechanic |
| u0089 | 4 | 8 | +0.08 | -0.09 | -1.31 | +0.23 | +1.73 | +0.22 | grandparent |
| u0137 | 6 | 8 | +0.00 | -1.22 | +1.91 | +1.54 | +0.89 | +0.97 | mechanic |
| u0139 | 4 | 8 | -0.04 | +0.43 | +0.02 | -0.49 | +1.12 | +2.17 | novelist |
| u0107 | 3 | 8 | -0.09 | +2.74 | -2.33 | -0.45 | +0.89 | +2.07 | novelist |
| u0121 | 7 | 9 | -0.17 | -0.24 | -0.21 | +0.20 | +1.01 | +2.35 | philosopher |
| u0135 | 6 | 9 | -0.28 | +0.16 | -1.01 | +0.87 | -0.15 | +0.50 | novelist |